

2.1.3 Proteins and enzymes

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) (i) the basic structure of an amino acid and the formation of peptide bonds (ii) the use of chromatography in the separation and identification of amino acids	To include calculation of retention (R_f) values. $R_f = \frac{\text{distance moved by the solute}}{\text{distance moved by the solvent}}$
(b) the molecular structure of globular proteins as illustrated by the structure of enzymes and haemoglobin	Practical work to be carried out in accordance with current CLEAPSS® guidelines with regards to the risks associated with the use of ninhydrin. Also see Section 5f. PAG6
(c) how the structure of globular proteins enable enzyme molecules to catalyse specific metabolic reactions	To include the role of tertiary structure in the specificity of the active site, the formation of enzyme substrate complexes and the lowering of the activation energy.
(d) (i) the factors affecting the rate of enzyme-catalysed reactions (ii) practical investigations into the factors affecting the rate of enzyme-catalysed reactions	<i>M0.1, M1.2, M1.3, M1.11, M3.1, M3.2, M3.3, M3.5, M3.6</i> PAG4 HSW3, HSW4, HSW5, HSW6, HSW8
(e) (i) the role of proteins in blood clotting, and blood clotting as an enzyme-controlled process (ii) the first-aid procedure to assist the blood clotting process and prevent excessive blood loss	To include the role of platelets, damaged tissue, thromboplastin, calcium ions, prothrombin, thrombin, fibrinogen and fibrin. HSW10, HSW11, HSW12
(f) the use of enzymes and inhibitors in medical diagnosis and treatment	To include diagnostic enzymes (e.g. blood amylase and LDH) and enzymes and inhibitors used in medical treatment (e.g. streptokinase, aspirin and warfarin).

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(g) the donation of blood, and the types and uses of stored blood products.	To include blood groups and a consideration of the other issues affecting blood donation AND the collection and use of whole blood, leuco-depleted blood, packed red cells, platelets, clotting factors and plasma. HSW9, HSW10, HSW11, HSW12